

Project brief

Name: SC Coverage Report Agent for Snipes / Nike order coverage

The agent will:

- 1. extract mock orderbook data from a database;
- 2. analyze product coverage by season and requested month;
- 3. calculate booked, available, open-order and delay coverage;
- 4. generate a structured report;
- 5. publish the result to Google Sheets;
- 6. email the report link to the user.

Business problem

Supply-chain and account teams need to understand whether future Nike/Jordan demand for a retailer is sufficiently covered. In a manual workflow, this usually requires pulling orderbook data, filtering statuses, building Excel pivots, checking delay buckets, formatting a report and sending it manually.

The agent removes this manual process and creates a repeatable autonomous workflow.

Core business questions

How much demand is already booked or shipped?	Booked/Shipped volume/value and percentage
How much stock is available to cover orders?	Available volume/value and coverage percentage
How much remains open?	Open-order volume/value by season and month
Where are the delivery risks?	Open orders split into Early/On Time, +1 week, +2 weeks, +3 weeks, +4 weeks or later
Which season needs attention?	Coverage by SP/SU/FA/HO season

Google Sheets output Tabs: 1. Executive Summary 2. Coverage by Season	Success criteria - Report runs without manual intervention. - Supabase totals reconcile to report totals. - Difference is no more than 0.01 for monetary values. - Google Sheet is created successfully. - Notification email contains a working report URL. - Failures are logged and returned to N8N.
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WORKFLOW OUTLINE

N8N Manual/Webhook Trigger



Python Agent Runner



LangGraph workflow

1. Validate request
2. Retrieve reporting rules from Pinecone
3. ReAct research agent selects and calls tools
4. Extract order data from Supabase
5. Validate and transform data
6. Calculate seasonal coverage
7. Generate report tables and commentary
8. Publish to Google Sheets
9. Send Gmail notification



Return run status and report URL to N8N

sc-coverage-agent/

- ├─ README.md
- ├─ requirements.txt
- ├─ .env.example
- ├─ .gitignore
- ├─ data/
 - ├─ sample_orderbook.csv
 - ├─ knowledge/
 - ├─ coverage_rules.md
 - ├─ data_dictionary.md
 - └─ validation_rules.md
- ├─ src/
 - ├─ config.py
 - ├─ state.py
 - ├─ graph.py
 - ├─ agent.py
 - ├─ main.py
 - ├─ tools/
 - ├─ supabase_tool.py
 - ├─ pinecone_tool.py
 - ├─ calculation_tool.py
 - ├─ sheets_tool.py
 - └─ gmail_tool.py
 - ├─ services/
 - ├─ transformations.py
 - ├─ validations.py
 - └─ report_builder.py
 - ├─ prompts/
 - └─ report_commentary.txt
- ├─ scripts/
 - ├─ load_supabase.py
 - └─ ingest_pinecone.py
- ├─ tests/
 - ├─ test_transformations.py
 - ├─ test_validations.py
 - └─ test_graph.py
- ├─ n8n/
 - └─ coverage_workflow.json
- └─ docs/
 - ├─ architecture.png
 - └─ demo-script.md

Stages	Focus Area	Tasks	Required Output
1	Scope, data model and setup	Confirm report definition and coverage formulas. Inspect ORDERBOOK, Select MVP fields. Create GitHub repository and Python environment. Create Supabase project and orderbook table. Load cleaned mock data. Define input/output contracts.	Repository, architecture draft, Supabase table, sample SQL query
2	RAG and deterministic reporting	Create reporting-rules documents. Create Pinecone index and ingestion script. Test retrieval. Build Python extraction, transformation and aggregation functions. Reproduce one season from the Excel report. Add reconciliation tests.	Working Supabase extraction, Pinecone retrieval, local report JSON
3	Tools, ReAct and LangGraph	Wrap Supabase, Pinecone, calculation, Sheets and Gmail functions as tools. Create ReAct agent. Define LangGraph state and nodes. Add routing, retries and validation. Run locally using mock publishing tools first.	End-to-end local LangGraph run
4	Google APIs and N8N	Connect Google Sheets and Gmail APIs. Create and format report tabs. Send notification email. Build N8N workflow with manual/webhook trigger, Execute Command and failure branch. Test complete workflow.	Google Sheet URL, notification email, exported N8N JSON
5	Testing, documentation and presentation	Run at least three test scenarios. Save generated report examples. Complete README, architecture diagram and tool descriptions. Add screenshots. Record demo. Clean secrets and finalize GitHub history.	Submission-ready repository and demonstration